

I. Introduction:

This report details a database system design, implementations, and sample queries for Savannah tourism book, with the primary goal being to take a set of semantic rules that describes a detailed list of Savannah attractions, businesses, hotels, and restaurants and convert them into a normalized universal relation database structure. The first part of this document establishes functional dependencies (FD) via twenty-six semantic rules, visualizes them in a Universal Relation Diagram, and then systematically decomposes the data through First (1NF), Second (2NF), Third (3NF), Boyce-Codd (BCNF), and Fourth (4NF) Normal Form. The second part of this document showcases the implementation of the database into a MySQL client, a database management software, including all INSERT statements and the completed relation it outputs. This document then concludes with two views of this database, a final set of seventeen queries, the results of said queries, and an appendix of meeting minutes with the database admin team.

II. Semantic Rules:

1. Each Book has an ISBN, title, publisher, edition number, year of publication, and a number of pages.
2. Each Book has several Attractions.
3. Each Book has several Businesses.
4. Each Book has several Hotels.
5. Each Book has several Restaurants.
6. Each Attraction has several Emergency Agencies.
7. Each Attraction has several Savannah area attractions.
8. Each Attraction has several Tybee Island area attractions.
9. Each Attraction has several Historic Cemeteries.
10. Each Attraction has several Historic Churches.
11. Each Attraction has several Historic Squares.
12. Each Attraction has several Public Parks.
13. Each Attraction has several Events.
14. Each Attraction has several Art attractions.
15. Each Attraction has several Sports attractions.
16. Each Attraction has several Universities.
17. Each Emergency Agency has a name and phone number.
18. Each Savannah area attraction has a name, location, phone number, website, and description.
19. Each Tybee Island area attraction has a name, location, phone number, website, and description.
20. Each Historic Cemetery has a name, location, year established, type, and description.
21. Each Historic Church has a name, phone number, location, denomination, website, and description.

22. Each Historic Square has a name, a location, a designer, a design year, architecture style, and description.
23. Each Public Park has a name, location, municipality, and description.
24. Each Calendar Event has a name, phone number, location, website, start date, and description.
25. Each Art attraction has a name, phone number, location, website, and type.
26. Each Sports attraction has several golf courses.
27. Each Sports attraction has several tennis courses.
28. Each Sports attraction has several boat ramps.
29. Each Sports attraction has several fishing piers.
30. Each Golf Course has a name, phone number, location, website, and description.
31. Each Tennis Course has a name, phone number, location, and description.
32. Each Boating Ramp has a name and location.
33. Each Fishing Pier has a name and location.
34. Each University has a name, phone number, email address, location, website, type, and description.
35. Each University has several colleges.
36. Each University has several programs.
37. Each University has several departments.
38. Each department in a university has a name and phone number.
39. Each Business has several Medical Establishments.
40. Each Business has several Pest Control Companies.
41. Each Business has several Plumbers.
42. Each Medical Establishment has a name, a specialty, a description, services offered, a location, hours of operation, website(s), and phone number(s).
43. Each Medical Establishment has several doctors.
44. Each Doctor has a name.
45. Each Doctor has several degrees.
46. Each Pest Control Company has a name, location, phone number, operating hours, specialty, payment method accepted, and description.
47. Each Plumber has a name, location, phone number, operating hours, specialty, payment method accepted, and description.
48. Each Hotel has a name, phone number, location, website, price range, booking restrictions, amenities, payment method accepted, historic area, description, and cancellation restrictions.
49. Each Restaurant has a name, phone number, location, website, food speciality, may or may not accept reservations, and description.